

Sahara Al-Madi

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PROFESSIONAL PROFILE

Computational linguist specializing in multilingual NLP, AI governance, and linguistic security. Founder of the Linguistic Security Institute (LSI), an open source research initiative focused on language technologies, AI governance, and linguistic security. Experience spans multilingual AI data operations, LLM evaluation, and quality assurance in industry, alongside research in model bias, dialect variation, and language technologies.

EDUCATION

Georgia Institute of Technology

M.S. Computer Science (Prospective Spring 2027)

University of California, Los Angeles (UCLA)

B.A. Applied Linguistics

PROFESSIONAL EXPERIENCE

Google (via TEKsystems)

AI Data Operations Consultant | May 2026–Present

- Support multilingual AI systems through data operations, annotation, localization, and quality assurance for enterprise-scale AI development.
- Collaborate with engineering teams to improve multilingual data quality, consistency, and model reliability.
- Execute high-priority production data updates with 100% on-time delivery.
- Contribute to evaluation workflows supporting multilingual large language models.
- Delivery Quality Reports to client and stakeholders

Key Areas

Multilingual AI • Golden Label Sets • Data Operations • Localization • Quality Assurance • LLM Evaluation

SoundHound AI (via Upwork)

AI Data Operations Consultant | October 2024– May 2026

- Led data operations for a multilingual conversational AI project, managing a team of contractors and ensuring quality and delivery milestones were met.
- Developed QA frameworks and audit processes that contributed to a measurable improvement in data integrity.
- Worked with the engineering team to identify and address annotation inconsistencies, supporting model performance across English and Spanish datasets.
- Created documentation and handover materials to support project continuity..

Key Areas

Voice AI • QA Frameworks • Data Annotation • Localization • Project Leadership

Linguistic Security Institute (LSI)

Founder & Lead Researcher | September 2025–Present

- Founded an independent research initiative advancing Linguistic Security as a framework for evaluating cultural, dialectal, and semantic integrity in AI systems.
- Develop open-source prototypes and benchmarking concepts for multilingual bias auditing, with an emphasis on Arabic and Spanish language technologies.
- Produce educational resources introducing linguistic security concepts to developers, researchers, and students.
- Develop governance proposals and risk assessment frameworks addressing language integrity, data sovereignty, and AI accountability.

National Museum of Language

Digital Strategy Lead | January 2020– August 2025

- Led multilingual digital strategy, increasing organic web traffic by 62%.
- Managed grant writing initiatives and cross-functional projects.
- Participated in board meetings with educators and government-affiliated stakeholders.
- Directed multilingual transcription, translation, and interpretation projects in English, Spanish, and Arabic.

RESEARCH & PUBLICATIONS

Arabic Polarization Detection Across 10+ Large Language Models (Published)

SemEval Task 9 at ACL 2026- Arabic Polarization Detection

Second-author, Error Analysis Lead

- Led error analysis across more than ten multilingual large language models, including DeepSeek, Qwen, Gemma, Llama, and Fanar.
- Investigated dialectal bias, hallucinations, Clever Hans effects, and gold-label inconsistencies.
- Contributed to published system paper describing comparative system analysis spanning generative, retrieval-augmented, and discriminative approaches.

BAREC 2026 Shared Task at EMNLP – Arabic Readability Assessment (Accepted)

Solo author, Presenting Researcher

- Built an ensemble model combining AraBERT, AraModernBERT, and CAMELBERT with SAMER lexicon features for fine-grained Arabic readability classification (19 levels).
- Achieved 1st place in the Constrained Track and 3rd place in the Open Track with a Quadratic Weighted Kappa (QWK) score of 81.7
- Ranked #1 in exact accuracy (50.6%), #1 in mean absolute error (1.0), and #1 in Acc 3 (77.2%), outperforming all submitted systems in precision metrics.
- Pending published system paper detailing methodologies, including ensemble learning, heuristic adjustments, and SAMER lexicon integration.
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Technology for Second Language Learning 2026

Linguistic Security Framework for Endangered Languages

Solo author, Presenting Researcher

INVITED TALKS & PRESENTATIONS

- **Tech Intersections (Northeastern University Oakland)** | January 2026
Linguistic Security and Marginalized Communities
- **GEO Indigenous Summit (Space4Innovation)** | March 2026
Who Speaks for the Data? A Linguist's Lens on Listening to Land and Language presented at an international conservation and governance conference
- **BSides San Diego** | April 2026
Polyglot Poisoning: Bypassing AI Guardrails via the Linguistic Supply Chain presented at an international Cybersecurity conference
- **TSL 2026** | October 2026
Linguistic Security and Endangered Languages to be presented at the Technology and Second Language Learning Conference
- **DEATHCon 2026** | November 2026
Polyglot Poisoning Workshop at an international Cybersecurity conference

OPEN SOURCE PROJECTS

Linguistic Firewall: Polyglot Poisoning

- Open-source Python toolkit for detecting multilingual prompt injection and polyglot poisoning vulnerabilities. Presented at BSides San Diego (2026).

Mitote

- Retrieval-Augmented Generation (RAG) text-to-speech prototype supporting Nahuatl-aware Spanish pronunciation for Indigenous language preservation.

CLI Linter for Language

- Early-stage command-line tool for identifying hard-coded English text and missing multilingual or dialect support in software projects.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy), SQL, Git, GitHub, Bash

AI & NLP: Large Language Model Evaluation, Error Analysis, Retrieval-Augmented Generation (RAG), Prompt Engineering, Zero-Shot Learning, Fine-Tuning Concepts, Bias Analysis

Data & Quality Assurance: Data Annotation, Localization, Gold Label Audits, Quality Assurance, Dataset Validation

Linguistics: Phonetics, Phonology, Syntax, Pragmatics, Dialectology, IPA, SAMPA

Languages: English (Native), Spanish (Native), Arabic (Reading, Writing, Linguistic Analysis), Italian (Intermediate)